

Effect of Six-weeks Progressive Resistance Training on Horizontal and Vertical Jumping ability among Volleyball Trainees

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ABSTRACT

This paper aims to determine the effect of resistance training intervention programme on women volleyball players' horizontal and vertical jumping ability. It was an experimental study, and twenty women volleyball players with a national school participation level were selected as the study sample. The age of the players ranged from fifteen to eighteen years. A pretest-posttest quasi-experimental design was used for the research. The players were exposed to a six-week progressive resistance training programme comprising squat, bench press, overhead press, deadlift, and barbell row. Tools for measuring horizontal and vertical jumping ability were the sargent jump test, approach spike jump test, three forward jumps tests, and standing broad jump test. After analysis of the data, it was found that the weight training intervention programme significantly affected and increased women volleyball players' horizontal and vertical jump abilities. Therefore, it is recommended that a progressive resistance programme must be applied by sport-strength trainers, coaches, players, and physical education teachers for the development of horizontal and vertical jumping abilities.

KEYWORDS: Weight-Training, Horizontal and Vertical Jumping Ability, Peak Performance and Explosive Strength.

INTRODUCTION

Considering the competitive nature of volleyball, sports scientists are always searching for new ways of improving performance (Chander, 2010). There are four variations apart from traditional

volleyball: beach volleyball, snow volleyball, and sitting volleyball in the Paralympics. However, the present research focuses on conventional volleyball. It consists of fundamental

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elements of volleyball, such as service, service reception, set, attack, block, floor defense, and counter-attack. Traditional articles explore the concept of sports training. Most of them are basic physiology, emphasize various exercises, and suggest a few training methods. Chander (2018) argued that arrangement is rarely discussed, and the structuring of resistance training is seldom indicated. The objective of resistance training must be to keep volleyball players ready for competition, polishing skills, and psychological readiness. To achieve the best results, there is a need among volleyball players to have volleyball-specific variations in training. Bompá and Buzzichelli (2019) have also discussed periodization in structuring a strength training program for athletes in various sports. It includes the planning of a training program for getting peak performance. They recommended half squats during the specific preparation phase. Further, they argued the need for strength development to power-generating capacity during the late preparatory phase.

Explosive strength is crucial in volleyball because it allows players to generate a powerful and quick burst of energy for jumping, spiking, blocking, and diving. These explosive movements require high force production in a short

time, so developing explosive strength is essential for volleyball players. In volleyball, explosive strength is significant for spiking, which is one of the most important offensive skills in the game. To execute a successful spike, a player needs to generate a high amount of force to jump high quickly and hit the ball with power and accuracy. In addition to spiking, explosive strength is also crucial for blocking. A blocker needs to be able to quickly jump up and reach over the net to stop the opponent's attack. Developing explosive strength in the legs and core is vital to achieving this. Overall, explosive strength is essential for any volleyball player who wants to perform at a high level. By training for explosive strength, players can improve their vertical jump, speed, agility, and overall power, which can translate into better performance on the court.

Developing a volleyball-specific combination of strength and power before the competitive phase is necessary because this forms the foundation of volleyball player performance. The critical element in resistance training for volleyball players is establishing a sequence of training phases for developing strength. Resistance training is more than just lifting weights every day. One requires having a specific target for particular training phases. Many

scholars create the role of strength with performance (Hardee, Lawrence, Zwetsloot, Triplett, Utter, and McBride, 2013; Contreras, Vigotsky, Schoenfeld, Beardsley, McMaster, Reyneke, and Cronin, 2017; Borràs, Balius, Drobnič, and Galilea, 2011; Cardoso Marques, Gonzalez-Badillo, and Kluka, 2006; Chander, and Mukherjee, 2012). The development of several types of strength in a specific sequence is required for achieving high levels of power and muscular endurance for volleyball playing ability, along with manipulation of load, force application, and repetitions during a workout, as well as the loading patterns in each phase, to create the specific adaptations for reaching peak performance.

Resistance training is used to improve all types of strength among volleyball players, i.e., maximum strength, explosive strength (power), strength endurance, or sometimes terminology also used as static strength, dynamic strength, and relative strength. Jumping ability is always the primary focus in Volleyball training and competition; hence, sizeable spaces are allocated for its improvement and capitalize on the player's jump height. Considering the need for jumping ability in volleyball, the research aims to

determine the effect of resistance training intervention programme on women volleyball players' horizontal and vertical jumping ability.

METHODOLOGY

This research was an experimental study. Twenty volleyball women players were selected as subjects whose participation level was national. The age of the subjects ranged from fifteen to eighteen years, with a minimum of four years of training history. A pretest-posttest quasi-experimental design was used for the research. The players were exposed to a six-week external resistance training programme comprising squat, bench press, overhead press, dead lift, and barbell row. The six weeks schedule of the self-designed intervention programme is depicted in table 1:

The resistance training programme was started with 75% weight of maximum strength and after every two weeks, there was an increase of 5% in resistance/ load volume. The volleyball game demands the jumping ability in horizontal and vertical directions and sometimes successive jumps, one after the other. Keeping in view the requirements and demands of the volleyball playing ability, the following tools were selected:

Table 1: The weekly schedule for the intervention programme

Six Weeks External Resistance Training Intervention Programme			
Week-1			
Day	Exercise Sets/Reps	Exercise Sets/Reps	Exercise Sets/Reps
Monday	Squat 6×6	Bench-Press 6×6	Barbell Row 6×6
Wednesday	Squat 6×6	Overhead Press 6×6	Dead Lift 1×6
Friday	Squat 6×6	Bench Press 6×6	Barbell Row 6×6
Week-2			
Monday	Squat 6×6	Overhead Press 6×6	Dead Lift 1×6
Wednesday	Squat 6×6	Bench Press 6×6	Barbell Row 6×6
Friday	Squat 6×6	Overhead Press 6×6	Dead Lift 1×6
Week-3			
Monday	Squat 6×6	Bench-Press 6×6	Barbell Row 6×6
Wednesday	Squat 6×6	Overhead Press 6×6	Dead Lift 1×6
Friday	Squat 6×6	Bench Press 6×6	Barbell Row 6×6
Week 4			
Monday	Squat 6×6	Overhead Press 6×6	Dead Lift 1×6
Wednesday	Squat 6×6	Bench Press 6×6	Barbell Row 6×6
Friday	Squat 6×6	Overhead Press 6×6	Dead Lift 1×6
Week 5			
Monday	Squat 6×6	Bench-Press 6×6	Barbell Row 6×6
Wednesday	Squat 6×6	Overhead Press 6×6	Dead Lift 1×6
Friday	Squat 6×6	Bench Press 6×6	Barbell Row 6×6
Week-6			
Monday	Squat 6×6	Overhead Press 6×6	Dead Lift 1×6
Wednesday	Squat 6×6	Bench Press 6×6	Barbell Row 6×6
Friday	Squat 6×6	Overhead Press 6×6	Dead Lift 1×6

1. **The Three Forward Jumps Test** or triple hop test (Hamilton, Shultz, Schmitz & Perrin, 2008) was used to measure an athlete's explosive leg power. Distance from starting and the landing point of the third jump in centimeters, and the best jump out of three attempts was treated as a score.

2. **The Standing Broad Jump Test** (ASC protocol for assessment of inertial strength ASC, 2000) was used to measure an athlete's explosive leg power and

lower body strength. The athletes were asked to jump forward as far as possible, landing with both feet together. Distance between the starting and landing point was measured in centimeters in three trials, and the best was treated as a score.

3. **The Sargent Jump Test** (Berry L. Johnson & Jack K. Nelson, 1982) was used to measure an athlete's explosive leg power and lower body strength in direction of vertical jump. The distance between standing reach and the highest

point with the jump was marked as a score in centimeters out of the three attempts.

4. **Approach Spike Jump** (Canadian Volleyball Association Physical test, 1978) aims to measure leg strength vertically with arm swing and approach. The subject was to stand with one side towards a wall, take two to three steps

with arm swings, and touch her finger marked with lime powder as high as possible nearest the wall-the number of centimeters between the approach spike jump reach and the ground level as the score. Three successive trials were given, and the best trial was recorded as the score.

RESULTS AND DISCUSSION

Table 2: Comparing the Pre- and Post-test of Muscular Strength Test

Variables	Tests	Stage	N	Mean	Std. Deviation	Std. Error Mean
Horizontal Jumping Ability	Three Forward Jumps Test	Pre-test	20	428.50	67.30	15.05
		Post-test	20	443.35	67.71	15.14
	Standing Broad Jump Test	Pre-test	20	172.15	21.19	4.73
		Post-test	20	179.55	21.48	4.80
Vertical Jumping Ability	Sargent Jump Test	Pre-test	20	31.10	6.23	1.39
		Post-test	20	36.95	6.33	1.41
	Approach Spike Jump Test	Pre-test	20	238.65	7.06	1.57
		Post-test	20	249.05	7.93	1.77

Note: Test performance score in centimeters was used as a measuring unit

Table (2) shows the results of paired sample statistics of the horizontal jumping ability of women volleyball players. The mean scores and standard deviation for a pre-test of three forward jumps tests are 428.50 and 67.30, with a 15.05 standard error of the mean. The mean and standard deviation for the three forward jumps test post-test is 443.35 and 67.71, with a 15.14 standard error of the mean. The mean scores and standard deviation for the standing broad jump test's pre-test are 172.15 and 21.19, with a 4.73 standard error of the mean. The mean and standard deviation for the

standing broad jump test post-test is 179.55 and 21.48, with a 4.80 standard error of the mean.

Similarly, Table (2) also highlights the vertical jumping ability scores of women volleyball players. The mean scores and standard deviation for a pre-test of the sargent jump test are 31.10 and 6.23, with a 1.39 standard error of the mean. The mean and standard deviation for the post-test of the sargent jump test is 36.95 and 6.33, with a 1.41 standard error of the mean. The mean scores and standard deviation for a pre-test of the approach spike jump test are 238.65 and

7.06, with a 1.57 standard error of the mean. The mean and standard deviation for the approach spike jump test post-test

are 249.05 and 7.93, with a 1.77 standard error of the mean.

Table 3: Pearson's Correlations

Pair	N	Pearson's Correlation	Sig. Level
Three Forward Jumps -Pre-test & Three Forward Jumps-Post-test	20	.999	p<.000
Standing Broad Jump -Pre-test & Standing Broad Jump-Post-test	20	.997	p<.000
Sargent Jump- Pre-test & Sargent Jump-Post-test	20	.977	p<.000
Approach Spike Jump -Pre-test & Approach Spike Jump-Post-test	20	.947	p<.000

Table (3) concluded that the correlation between pre-test & post-test scores of the horizontal and vertical jumping ability of

women volleyball players and all pairs was statistically highly significant.

Table 4: Paired Sample "t" Test

Pair	Paired Differences					t-value	df	Sig. (2-tailed)
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
				Lower	Upper			
Three Forward Jumps-Pre-test - Three Forward Jumps-Post-test	-14.850	3.232	.722	-16.362	-13.337	-20.544	19	p<0.000
Standing Broad Jump-Pre-test - Standing Broad Jump-Post-test	-7.400	1.759	.393	-8.223	-6.576	-18.812	19	p<0.000
Sargent Jump - Pre-test - Sargent Jump - Post-test	-5.850	1.348	.301	-6.481	-5.218	-19.401	19	p<0.000
Approach Spike Jump -Pre-test - Approach Spike Jump-Post-test	-10.400	2.583	.577	-11.609	-9.190	-18.004	19	p<0.000

From Table 4, it is indicated that the mean differences between pre-test & post-test scores of the horizontal and vertical jumping ability of women volleyball players, as per table scores it, is found to be statistically significant. The t-value of all pairs, namely three forward jumps pre-test versus three forward jumps post-test, standing broad jump-pre-test versus standing broad jump-post-test, sergeant jump pre-test versus sergeant jump post-test, approach spike jump pre-test versus approach spike jump-post-test is greater than table value 1.729, with the degree of freedom 19 at 0.05 level significance level.

This shows the overall effectiveness of the six-week resistance training intervention programme on women volleyball players' horizontal and vertical jumping ability. Jumping ability in volleyball is a critical component in spiking and blocking and is also used as a talent identification tool to examine/evaluate explosive strength (Fry, Kraemer, Weseman, Conroy, Gordon, Hoffman, Maresh, 1991). Overall, resistance training can be a valuable tool for improving explosive strength in volleyball players. By incorporating these exercises into their training programs, players can improve their performance on the court and take their game to the next level. The Three Forward Jumps Test can be used to track an athlete's progress over time and can be helpful in assessing the effectiveness of

training programs aimed at improving explosive leg power.

CONCLUSION

In volleyball, the playing ability depends on highly functional physical qualities, and one of such qualities is jumping ability along with power, anticipation, and decisive skills. One must respond to the various trajectory balls and opponents and cope with teammates. It is an intermittent sport, and jumping ability is an essential part of the spike, block, and jumping services. This research revealed that resistance training intervention programme significantly affects the horizontal and vertical jumping ability of women volleyball players. Combining it with volleyball training mesocycle sessions is recommended to develop overall volleyball playing ability. The players also reported that after the intervention programme, they felt and perceived more stay in the air during jumping. Considering the encouraging results of this research, it is recommended that a bigger sample size be used to generalize results. This research is helpful for volleyball sport-strength trainers, coaches, players, and physical education teachers for practical application by increasing conceptual knowledge of resistance training and its foundation, which may produce better performance. The best sports performance occurs because of research and intelligent planning.

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